Page No:
Date.

Final exam

(1) Q1 theory

(1)

mean salary :- $3575.1e
median salary :- $3400.0
mode salary :- ₹2000

(2)

emp salaries ₹20,000, 30,000, 40,000

Range = 40000 - 20,000
= ₹20,000

$$\text{mean} = \frac{20000 + 30000 + 40000}{3}$$

$$= 30000$$

Population varriance :=

Page No:
Date.

(1) Salary = 20,000, 30000, 40000, 50000 +30000

mean = $\frac{\text{sum of salaries}}{\text{Number of salaries}}$

$= \frac{20000 + 30000 + 40000 + 50000 + 30000}{5}$

$= 30000$

mode :- most frequently occuring value
= 30,000

(2) Data = 2, 4, 6

mean = (2 + 4 + 6) / 3 = 4

squared deviation

$(2-4)^2 = 4$
$(4-4)^2 = 0$
$(6-4)^2 = 4$

sample variance = (4 + 0 + 4) / (3 - 1)
= 8/2
= 4

Standard deviation
$= \sqrt{4}$
= 2.

(1) Probability

$$= \frac{\text{Number of promoted Employess}}{\text{Total employees}}$$

Total employee: 4000
promoted employe.: 1440

$$\text{probability} = \frac{1440}{4000} = 0.36$$

$$= 36.1\%$$

(2) Contingency table

Promation status / Department	NO	YES
Finance	999	176
HR	893	193
It	653	372
marketing	336	170
operation	307	997
Sales	512	978.

Page No:
Date.

Conditional Probability

P(Promotion | score > 80) = Promoted empl > 80 / All empl > 80

Promoted empl > 80 = 62

All Employee score > 80 = 84

= 0.7381

= 73.81%.

* Consider the first two employee vectors from dataset

A = (5, 44)

B = (5, 44)

1. Dot product

A · B = (a1 x b1) + (a2 x b2)

= (5 x 5) + (44 x 44)

= 25 + 2401

Dot Prod = 2426.

Page No:
Date:

5 Norm 1 of employee A

Norm 1 = |a1| + |a2|
= 5 + 49
= 54

6 Norm 2 of Employee A

Norm 2 = $\sqrt{(a1)^2+(a2)^2}$
= $\sqrt{(5)^2+(49)^2}$
= $\sqrt{25+2401}$
= $\sqrt{2426}$

Norm 2 = 49.2544

7 Angle between Two employee vector

$\theta = \cos^{-1}[(A.B) / (\|A\|_2 \times \|B\|_2)]$
= $\cos^{-1}(2426/(49.2544 \times 49.2544))$
= $\cos^{-1}(1)$

Angle = 0 degrees